

CoMn-Spinel Oxides as Supported Catalyst for Rocket-Grade Hydrogen Peroxide Decomposition

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Abstract: This work presents the application of a cobalt and manganese mixed oxides-based catalyst with spinel structure to the decomposition of the storable green monopropellant H_2O_2 90 wt%. The 5.4% $\text{Co}_{0.5}\text{Mn}_{2.5}\text{O}_4/\text{Al}_2\text{O}_3$ catalyst was tested in a series of continuous and pulsed 2 N thruster firings, yielding a fast and repetitive performance, spontaneously and completely decomposing H_2O_2 90 wt% without undergoing deactivation or fragmentation during the tests. A specific impulse of 104 s and a characteristic ve-

locity of 881 m/s were reached, so that the characteristic velocity efficiency amounted to 93.7%. The adiabatic decomposition temperature of the monopropellant (756 °C) was also reached, thus evidencing the effectiveness of the 5.4% $\text{Co}_{0.5}\text{Mn}_{2.5}\text{O}_4/\text{Al}_2\text{O}_3$ catalyst. Its excellent activity in the decomposition of H_2O_2 90 wt% was attributed to the presence of the redox pairs $\text{Mn}^{2+}/\text{Mn}^{3+}$ and $\text{Co}^{2+}/\text{Co}^{3+}$, therefore making it a feasible material for low and medium thrust propulsive systems.

Keywords: Spinel · Monopropellants · Catalysts · Hydrogen peroxide · Green propellants

1 Introduction

Artificial satellites depend on propulsive systems to compensate for orbit deviations caused mainly by gravitational and centrifugal forces. Monopropellant rocket engines are the most widely used ones for attitude control and satellite orbit correction. Currently, hydrazine (N_2H_4) and its derivatives like monomethyl hydrazine ($\text{CH}_3(\text{NH})\text{NH}_2$) are the monopropellants of choice in operating satellites, due to their remarkable stability and performance. In contrast, their toxicity and carcinogenicity have motivated the search for green monopropellants, capable of yielding good propulsive performance without the drawbacks inherent to N_2H_4 [1].

Alternative green monopropellants, such as HAN (hydroxylammonium nitrate), ADN (ammonium dinitramide) and HNF (hydrazinium nitroformate), have been extensively studied, but their advantages have not yet reached the legacy technology status of hydrazine and its derivatives [1]. In its turn, nitrous oxide (N_2O) features a number of advantageous aspects such as self-pressurization and environmental friendliness, but develops quite high decomposition temperatures (1873 K), drastically reducing, as a result, catalyst life [2]. Circumventing this feature requires catalysts capable of repeatedly withstanding elevated temperatures without significative microstructure alterations and with no assistance of typical low-thrust refrigerated chambers.

Rocket-grade hydrogen peroxide (H_2O_2 90 wt%) is considered a strong candidate to replace N_2H_4 in low and medium thrust propulsive systems, because it does not show the disadvantage of a remarkably high decomposition temperature, neither toxic nor carcinogenic effects [3]. Al-

though the specific impulse of N_2H_4 is about 20% higher than that of H_2O_2 90 wt%, it yields a density specific impulse roughly 10% lower. This stems from the 40% higher density provided by H_2O_2 90 wt% [4]. This feature is of paramount importance for volume-restricted systems such as satellites. However, the challenge facing the use of H_2O_2 90 wt% as a monopropellant is the development of a catalyst that promotes its fast and efficient decomposition. In addition, the catalyst shall have high mechanical strength and long service life, being resistant to the deactivation caused by structural modifications of the active phase. The catalyst should also be able to withstand several thermal cycles imposed by typical space mission profiles and, if possible, provide a spontaneous start of the decomposition without the need for preheating the catalytic bed [5].

Noble metals and manganese oxides supported catalysts have been widely studied for the decomposition of

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H₂O₂ 90 wt%. The catalyst traditionally used in N₂H₄ based propulsive systems, 36% Ir/ γ -Al₂O₃, was tested with H₂O₂ 85 wt% [6]. However, the authors observed a relatively poor chamber temperature of 480 °C, below the decomposition temperature of this monopropellant. This was justified by a strong catalyst deactivation as a function of time. This deactivation may have been caused by the oxidation of the metal or the adsorption of stabilizers present in H₂O₂ 85 wt% on the catalyst surface, thus reducing the life of the material.

Other authors suggested the use of noble metal-based catalysts in the decomposition of concentrated hydrogen peroxide for propulsive systems. Sahara et al. [7] and Pasini et al. [8] studied the use of platinum supported on alumina pellets and verified that thermal stresses induced by the intense release of heat often caused the catalyst to break into fine particles. Torre et al. [9] loaded α -Al₂O₃ with 35 wt% Pt to solve this problem. Despite this large amount of active phase, the characteristic velocity (*c**) efficiency reached 94% and the decomposition temperature attained 93% of its theoretical value.

Rusek [10] compared the performance of alumina-supported platinum and manganese oxides in the decomposition of rocket-grade hydrogen peroxide. The results showed that the catalysts presented very similar activities. However, tetravalent manganese yielded a higher catalytic activity as compared to trivalent manganese. Recently, H₂O₂ 98 wt% decomposition was achieved with catalysts consisting of 1 wt% platinum and of 1.1 wt% manganese and cobalt oxides, supported on silica-doped alumina [11]. According to the authors, these catalysts performed well in a 22 N thruster, yielding a 95% characteristic velocity efficiency, although the catalytic support did not exhibit significant specific area (0.38 m²/g). This favorable *c** efficiency result required the catalytic bed to be preheated up to approximately 327 °C. For cold starts, *c** efficiency varied between 45 and 60%, thus evidencing the need for preheating, probably due to the low specific area of the catalyst. Moreover, as the calcination temperature of the active phase (700 °C) was below the adiabatic decomposition temperature of H₂O₂ 98 wt% (937 °C), transformations occurred in the active phase during the tests.

Maia et al. [12] investigated the decomposition of H₂O₂ 90 wt% by bulk catalysts based on Co, Mn and Al mixed oxides in a 2 N thruster. Although it also had a low specific area (6 m²/g) after calcination at 900 °C, the catalyst performed well in this reaction. An average specific impulse of 97 s was obtained. This value was acceptably close to 106 s, the CEA NASA predicted figure. On the contrary, the theoretical value of characteristic velocity was 940 m/s and the experimental average value was 822 m/s, corresponding to a *c** efficiency of 87%, which indicates incomplete catalytic decomposition.

In order to better explore the potential advantages of H₂O₂ 90 wt% as a well-suited monopropellant for low and medium thrust propulsive systems, particularly the ones re-

quired by satellites, it is necessary to develop tailor-made catalysts, capable of delivering good performance through an effective decomposition without preheating. In addition, these catalysts should withstand the stresses associated to repeated thermal cycles and be available at acceptable cost. Therefore, this work investigated the preparation of a spinel oxide (Mn_xCo_{3-x}O₄) supported on alumina grains, favoring the occurrence of the redox pairs Mn³⁺/Mn²⁺ and Co³⁺/Co²⁺, to be tested as catalyst on a 2 N thruster operating with H₂O₂ 90 wt%.

2 Experimental

2.1 H₂O₂ Concentration

H₂O₂ 70 wt% (Interox 70–10), supplied by Peróxidos do Brasil (Solvay Group), was concentrated up to 90 wt%. The concentrator consists of Ladden tubes where the H₂O₂ is injected on one side and dry hot air flows on the other side. Concentration occurs through drying of H₂O₂. H₂O₂ concentration was determined by permanganometry.

2.2 Catalyst Preparation and Characterization

A γ -alumina AxSorb™ (AxienS) was crushed and spheroidized into grains with a mesh size of 16–28 (0.59–1.00 mm). Subsequently, solutions of Co and Mn nitrates (Sigma-Aldrich) in the ratio of 0.5:2.5 Co:Mn were prepared, maintaining a concentration of 2 mol/L of the metal cations (Co²⁺ and Mn²⁺). The alumina, previously dried at 120 °C for 2 hours, was impregnated once with the nitrate solution until its wet point was reached. Finally, the catalyst was dried at 120 °C for 2 hours and calcined at 900 °C for further 2 hours.

Catalysts were characterized before and after the thruster firings. Specific surface area was determined through nitrogen adsorption and active phase dispersion was obtained by CO chemisorption with a Micromeritics ASAP 2020 C surface analyzer. Active phase content stemmed from Inductively Coupled Plasma Optical Emission Spectrometry (ICP-OES), performed in a Thermo Scientific model iCAP 7000. The crystalline phases in the prepared catalyst were identified through X-ray diffractometry (XRD), in a Shimadzu model XRD-6000. Finally, mechanical compressive stress was assessed according to ASTM D 7084–04 with a Shimadzu model AG–X Universal Machine with a 5 kN load cell.

2.3 Assessment of Thruster Performance

A thruster with 2 N theoretical thrust was used to determine the experimental parameters of the catalyst performance in the decomposition of H₂O₂ 90 wt%. This thrust-

er was made of 316 stainless steel and consisted basically of three independent modules: the injector, the chamber that housed the catalytic bed and the nozzle. The modules were interconnected through flanges, and the seal was made of a copper ring. Three temperature and two pressure sockets were installed, as shown in Figure 1.

The catalytic bed was 30 mm in length and 15 mm in diameter, the same dimensions used by Maia et al. [13]. It was loaded with 7.45 g of catalyst, restrained by two stainless steel screens, each one 0.5 mm thick. The nozzle had a throat area (A_t) of 3.6 mm² and an expansion ratio (ϵ) of 1.45, adjusted to provide optimum expansion to an atmospheric pressure of 96 kPa. The thruster was designed to operate under a 2 g/s mass flow rate of H₂O₂ 90 wt.% and a chamber pressure of 500 kPa. Assuming the frozen flow hypothesis, CEA NASA predicted a theoretical specific impulse of 106 s and a characteristic velocity of 940 m/s.

The experimental variables monitored were thrust, chamber pressure and catalytic bed temperature. They allowed the calculation of the specific impulse (I_{sp}) and of the characteristic velocity (c^*). The tests were performed on a portable test bench as shown in Figure 2.

3 Results and Discussion

3.1 Catalyst Preparation

Preliminary results regarding the effect of Co:Mn ratio and active phase content led to the synthesis of supported Mn_xCo_{3-x} catalyst with $x=2.5$, prepared from a single impregnation step. The catalyst was supported on AxorbTM alumina with a specific area of 287 m²/g. This catalytic support was impregnated with the active phase and calcined at

900 °C, reducing the surface area of the material to 78 m²/g. This calcination temperature was selected because it is slightly higher than the adiabatic decomposition temperature of H₂O₂ 90 wt%, thus preventing structural transformations of the catalyst during its operational life. The catalyst prepared containing the Co:Mn molar ratio of 0.5:2.5 showed a content of 0.75 wt% of Co and 3.56 wt% of Mn and could be represented as follows: 5.4% Co_{0.5}Mn_{2.5}O₄/Al₂O₃. It is noteworthy that higher contents of active phase were tested, but the density of active sites present on the catalyst surface tended to decrease as the active phase content increased, consequently reducing the catalyst activity. This suggested a low dispersion of the active phase on the support surface, probably due to the overlapping of the oxide particles during impregnation. It is worth mentioning that the catalyst's choice stemmed from preliminary drop tests carried out with materials of the Mn_xCo_{3-x} kind, where "x" ranged from 0 to 3 at 0.5 increments, all prepared in the same fashion described for the assessed catalyst. The tests consisted of bringing into contact H₂O₂ 70 wt% and a single grain of each of the materials. The drop tests were monitored with a FASTEC S100 model and showed that the Co_{0.5}Mn_{2.5}O₄/Al₂O₃ catalyst was the one that presented the shortest time to start the decomposition (induction time) of H₂O₂.

3.2 Catalyst Characterization

Figure 3 presents XRD patterns of the synthesized 5.4% Co_{0.5}Mn_{2.5}O₄/Al₂O₃ catalyst. The diffraction peaks for Co_{0.5}Mn_{2.5}O₄ could be indexed to the intermediate tetragonal spinel structure between cubic MnCo₂O₄ and tetragonal Mn₃O₄ (JCPDS no. 18-0408) [14]. The structure formed was

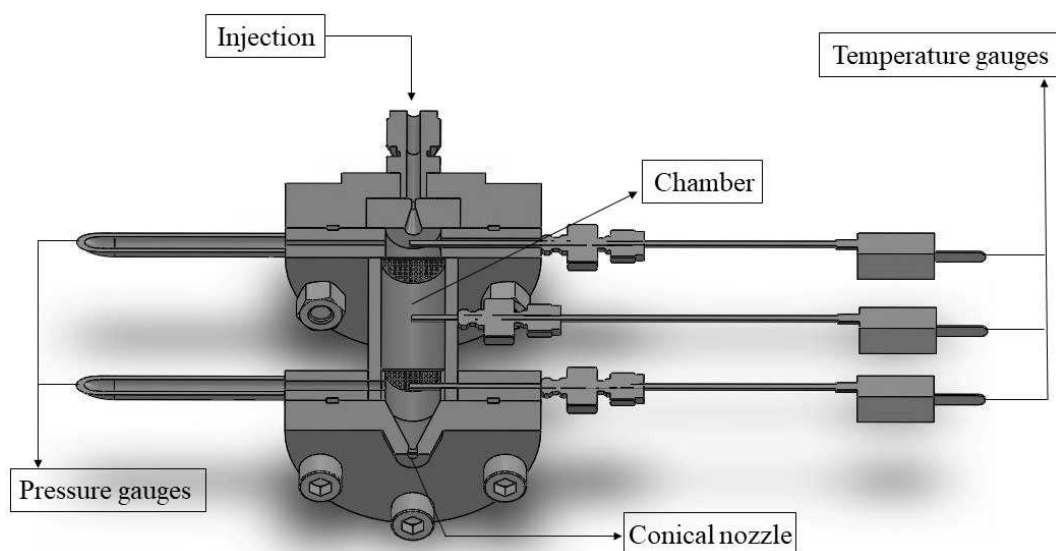


Figure 1. 2 N thruster schematics.

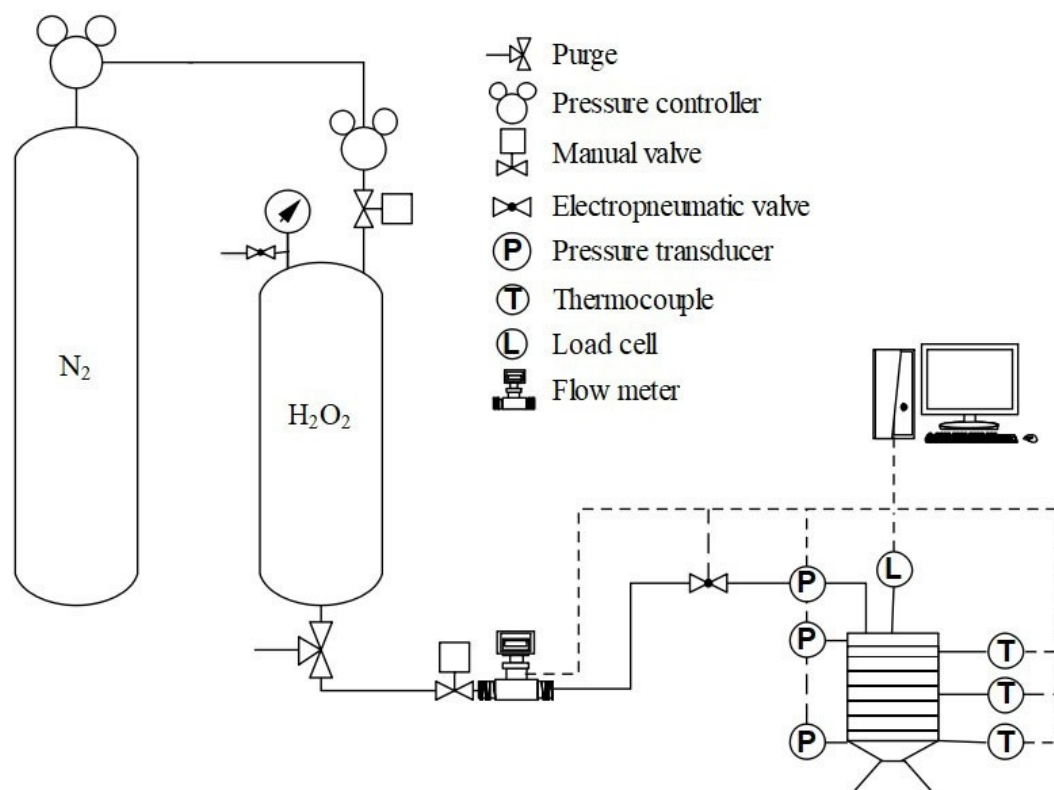


Figure 2. Schematic representation of the test stand.

a mixed valence spinel oxide with a cubic structure in which manganese and cobalt are distributed over both octahedral and tetrahedral sites. Furthermore, $\text{Co}_{0.5}\text{Mn}_{2.5}\text{O}_4$ showed no cobalt oxide peaks, indicating that all the cobalt compounds were converted to cobalt manganese spinel oxides, due to a strong interaction between cobalt and

manganese at the molecular level [15]. In addition, poorly crystallized phases for MnO_2 (JCPDS no. 42–1316) and γ -alumina (JCPDS no. 47–1292) were detected. The presence of γ -alumina indicates that the calcination time was not long enough to convert it to θ -alumina.

As for the mechanical assessment, a compressive stress of 1.77 MPa was obtained to generate 1% of fines with particle size lower than 0.42 mm in the catalyst samples. The literature points out stress values ranging from 0.7 to 3.22 MPa to produce such particulate size proportion, thus placing the synthesized catalyst within the practicable range [16].

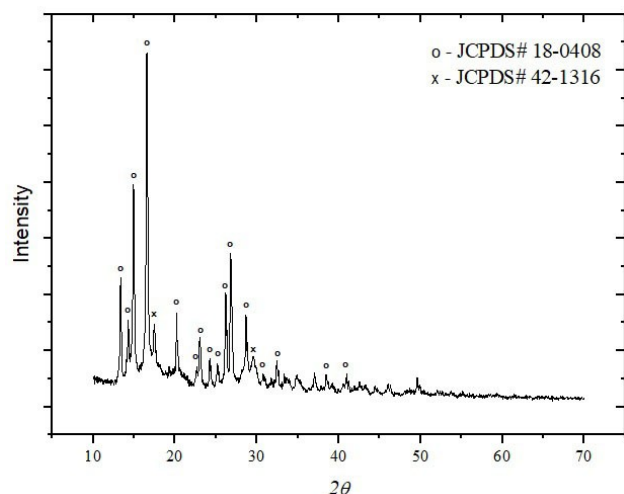


Figure 3. XRD patterns of the synthesized 5.4% $\text{Co}_{0.5}\text{Mn}_{2.5}\text{O}_4/\text{Al}_2\text{O}_3$ catalyst.

3.3 Thruster Firings

Figure 4 shows the thrust-time trace obtained in a continuous 30-seconds firing. The thruster loaded with the 5.4% $\text{Co}_{0.5}\text{Mn}_{2.5}\text{O}_4/\text{Al}_2\text{O}_3$ catalyst achieved an average thrust of 2.04 N, very close to the theoretical value (2 N) with quite satisfactory stability. The small low frequency instability observed in the trace is not unusual and may be caused by coupling between the chamber and the supply line pressures, known as chugging, which is common in propulsive systems involving fluid feeding [17]. However, no spikings were observed in all the tests performed.

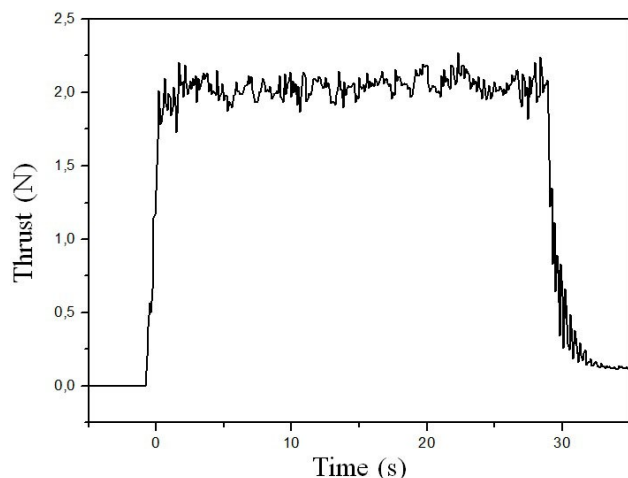


Figure 4. Thrust-time trace for the continuous 30-seconds test.

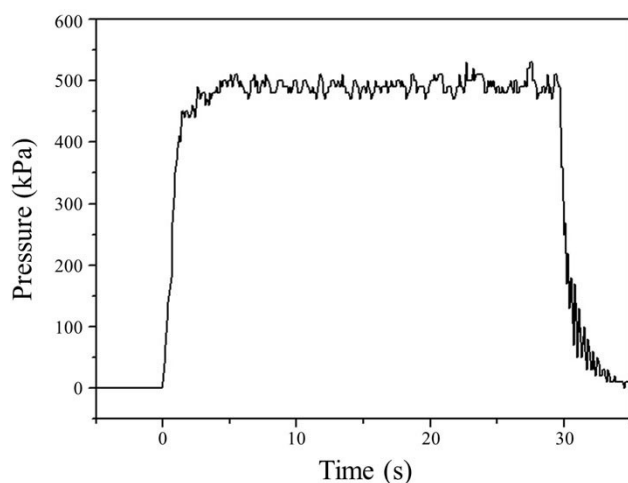


Figure 5. Pressure-time trace for the continuous 30-seconds test.

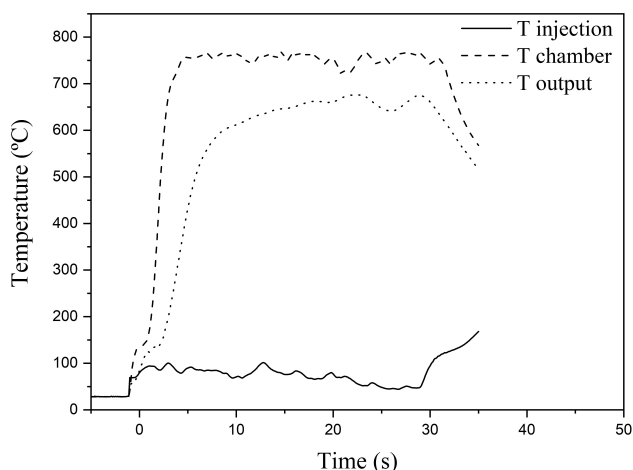


Figure 6. Temperature-time trace for the continuous 30-seconds test.

Figure 5 shows the chamber pressure-time trace during the 30-seconds continuous firing. The thruster achieved an average chamber pressure of 490 kPa, quite acceptable in view of the initial setting (500 kPa). The signal oscillation in the pressure curve is very similar to that shown in the thrust curve. Stable propulsive systems should operate within a $\pm 5\%$ pressure fluctuation around the mean chamber pressure [18]. A 6% oscillation was observed in this study, showing that this propulsive system is reasonably stable.

Figure 6 shows the temperature-time curves at the injection region, at the center of the catalytic bed, and at the catalytic bed outlet during the 30 seconds firing, conducted from a cold start, i.e. with the catalyst initially at room temperature. The average time to induce H_2O_2 90 wt% decomposition from cold starts, assessed through drop tests, was approximately 30 ms. This was possible due to the elevated catalyst specific surface ($78 \text{ m}^2/\text{g}$). Catalysts with a low specific area tend to have high ignition delays, thus requiring preheating of the catalytic bed [11].

In Figure 6, it can be observed that the injection temperature remains low throughout the test. In fact, this would be expected if part of the injected monopropellant, not yet decomposed, were able to cool the initial region of the catalytic bed. Coherently, at this region, H_2O_2 90 wt% remained almost entirely in the liquid phase and the temperature did not exceed the level of liquid/vapor coexistence. Measurements at the midpoint of the catalytic bed revealed the complete monopropellant decomposition at this sector as the adiabatic decomposition temperature of H_2O_2 90 wt% (756°C) was reached. The temperature at the catalytic bed outlet remained below the one measured at the center of the catalytic bed. This behavior may have been caused by a slight expansion of the gases after they left the catalytic bed confinement.

From the thrust and chamber pressure values obtained in the 30-seconds firing, it was possible to calculate the specific impulse (I_{sp}) and the characteristic velocity (c^*) of the system. The experimentally obtained I_{sp} (104 s) was slightly lower than the theoretical one (106 s), probably due to heat transfer losses between the thruster and the environment, once the average thrust agreed quite satisfactorily with its predicted value, yielding a relative error of 2%. The characteristic velocity allows to evaluate the catalyst's decomposition efficiency. The experimental c^* was 881 m/s, while the theoretical one was 940 m/s. According to Sutton and Biblarz [2], the experimental characteristic velocity should lie between 92 to 99.5% of its theoretical counterpart. Hence, the tested thruster achieved a characteristic velocity efficiency of 93.7%, thus suggesting, together with the catalytic bed temperature, that the decomposition of H_2O_2 90 wt% occurred completely.

For a better evaluation of the catalyst performance and stability, it was performed a 60-seconds continuous test and a pulsed test with 5-second shots at 5-second intervals. All these firings were carried out with the same original

load of catalyst. The thrust curves obtained are shown in Figures 7 and 8.

Figures 7 and 8 show that the catalyst exhibited good stability and showed no signs of deactivation over time. The pulsed test (Figure 8), performed before the continuous test, showed that the catalyst is capable of a fast and repetitive performance. These characteristics are very important for catalysts to be employed in satellite propulsive systems.

After the tests campaign, the thruster was disassembled, and the catalyst passed through a 0.59 mm sieve. The amount passing through this mesh was 0.04 g, corresponding to 0.54 % of fine particles. This percentage was very low, attesting the good mechanical strength of the catalyst. It is worth remembering that the cumulative thruster operation time, considering continuous and pulsed tests, using the same catalyst load, was 6 minutes, decomposing in this period approximately 720 g of H_2O_2 90 wt%.

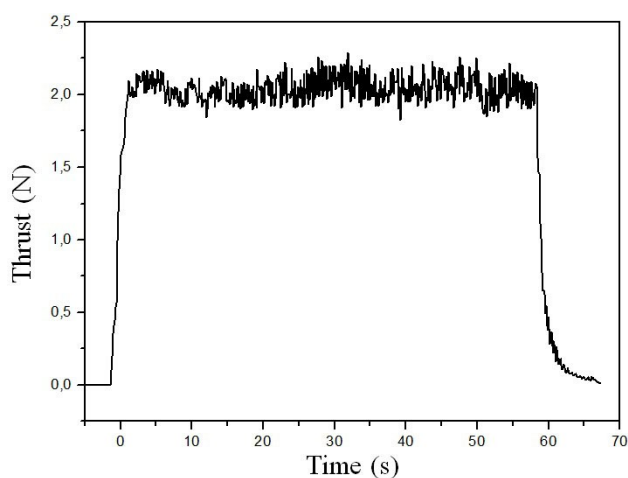


Figure 7. Thrust-time trace for the continuous 60-seconds test.

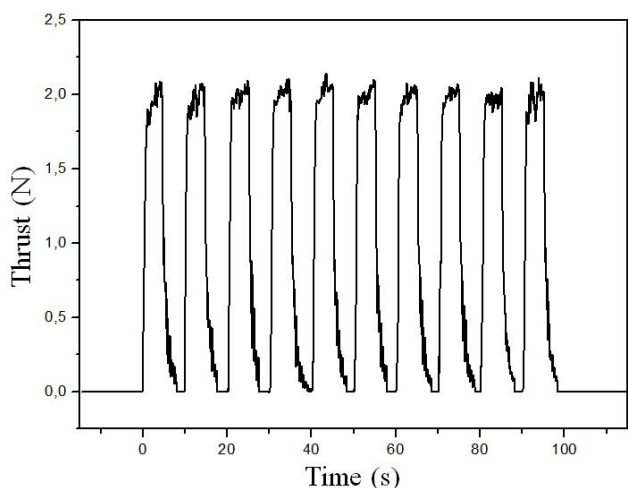


Figure 8. Thrust-time trace for the pulsed test.

The results showed that mixed manganese oxides-based catalysts were very active in the decomposition of H_2O_2 90 wt%, mainly the oxides MnO_2 and Mn_3O_4 . However, the addition of cobalt oxides promoted the formation of mixed oxides with different oxidation states on the support surface. This led to the formation of redox pairs $\text{Mn}^{2+}/\text{Mn}^{3+}$ and $\text{Co}^{2+}/\text{Co}^{3+}$, extremely active in the decomposition of H_2O_2 90 wt%. These atoms with different oxidation states, residing in the same lattice, allow electrons exchange interactions, thus providing the necessary electron-mobile environment for the surface redox activity [18].

The catalyst underwent further characterization in order to get a complete assessment of its performance. Inductively Coupled Plasma Optical Emission Spectrometry analysis did not detect differences in cobalt and manganese levels before and after the tests, and there was no active phase loss due to the thruster operation. This was attributed to the strong anchoring of the active phase on the support surface, probably caused by the electron density of the catalyst oxides. There were also no significant changes in the texture of the catalyst during the tests. At first the catalyst had a specific area of $78 \text{ m}^2/\text{g}$ and after testing $77 \text{ m}^2/\text{g}$. Such behavior was already expected since, in the preparation of the catalyst, a heat treatment was performed at temperatures higher than those reached in the thruster chamber during the decomposition of H_2O_2 90 wt%. The results from CO chemisorption showed that the active sites densities on the catalyst surface were the same before and after the thruster firings (around $95 \mu\text{mol}/\text{g}$). Finally, no changes were detected by XRD in the chemical state of the species found on the catalyst surface. This showed a good interaction of the active phase with the support, that is, even after being subjected to drastic temperature and pressure conditions, the catalyst's active phase remained unchanged.

4 Conclusion

The 5.4 % $\text{Co}_{0.5}\text{Mn}_{2.5}\text{O}_4/\text{Al}_2\text{O}_3$ catalyst was successfully prepared with a mesh size of 16–28 (0.59–1.00 mm), containing a Co:Mn molar ratio of 0.5:2.5 and a 0.75 wt% content of Co and 3.56 wt% content of Mn. The spontaneous decomposition of H_2O_2 90 wt% promoted by this catalyst was assessed in a series of continuous and pulsed 2 N thruster firings. An average specific impulse (I_{sp}) of 104 s was reached, very close to the theoretical value of 106 s. The average characteristic velocity amounted to 881 m/s, thus yielding a 93.7 % c^* efficiency. The adiabatic decomposition temperature of H_2O_2 90 wt% (756°C) was also reached, indicating that the monopropellant was completely decomposed. A tests campaign comprising continuous and pulsed firings suggested that the catalyst possessed good stability, showing no signs of deactivation after 6 minutes of thruster cumulative operation. The catalyst showed no morphological modifications, generating only 0.54 % of fines. Likewise,

structural changes in the catalyst were not observed, keeping the active phase intact after testing. The excellent catalyst activity in the decomposition of H_2O_2 90 wt% was attributed to the presence of the redox pairs $\text{Mn}^{2+}/\text{Mn}^{3+}$ and $\text{Co}^{2+}/\text{Co}^{3+}$. Therefore, the 5.4% $\text{Co}_{0.5}\text{Mn}_{2.5}\text{O}_4/\text{Al}_2\text{O}_3$ catalyst seems to meet the requirements for decomposing the storable green monopropellant H_2O_2 90 wt% in low and medium thrust propulsive systems.

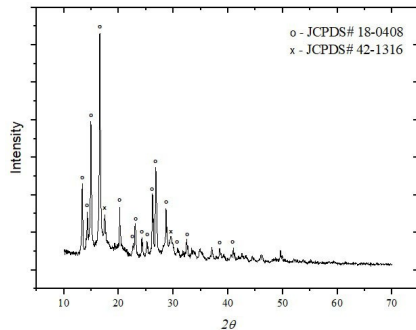
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1 – 8

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